

Lifestyle's Effect on Health Insurance Claims

By: James Christensen

Introduction

Health is the prerequisite for all human endeavors. Being healthy is something most of us take for granted, yet we are all intrinsically aware that it is constantly at risk. As of 2023, 92% of Americans had health insurance (public or private) (Keisler-Starkey, K. & Bunch, L. N.). This insurance is a tremendous financial burden for most. In fact, the average health insurance premium for family coverage in the US in 2024 was \$25,572. An increase of 24% since 2019 (KFF). These costs sum to an insurance market with a volume of \$318.4 billion dollars a year (Health Insurance - worldwide: Statista market forecast).

Since health insurance is both a necessity and expensive, it then begs the question, *is there a practical way for individuals to reduce how much their insurance company spends on them, and thereby reduce how much the individual spends on insurance?*

Dataset overview

To answer this question, we analyzed the [Medical Insurance Cost Prediction dataset](#) from Kaggle. This dataset includes information on 100,000 individuals including their demographics, socioeconomic status, health conditions, lifestyle factors, and medical expenditures. The dataset has a CC0: Public Domain License and was “compiled from publicly available health surveys, insurance research studies, and anonymized online healthcare data” (Thalla). It covers individuals around the globe from the years 2014 through 2024.

Factors and variables

To answer the question, we selected the `total_claims_paid` column to be our target variable (Y). As our predictor variables we selected `bmi`, `smoker`, `alcohol_freq`, `urban_rural`, `employment_status`, and `income`. `Bmi` is the bmi of the individual. `Smoker` indicates whether the individual actively smokes, formerly, or never has. `Alcohol_freq` was split into three levels. Individuals who occasionally drink, drink weekly, or drink daily. `Urban_rural` indicates whether the individual lived in an urban, suburban, or rural setting. `Employment_status` originally had four levels of employed, self-employed, unemployed, and retired. Since we wanted the effect of working on healthcare costs, we combined the employed and self-employed levels into a singular “employed” level. Finally, `income` is the annual income of the individual.

Data preprocessing

The only variable which had missing values was `alcohol_freq` and it had 30,083 of them. Since there are 100,000 values in the dataset, a possible option was to remove all of the missing value rows as there would still be substantial power in the model. In order to see if this was proper, we conducted two sample t-tests for difference in means between the values that were missing and the remaining data for numerical variables, and Chi-Square tests for difference in proportions for the categorical variables. Figure 1 displays the results of those tests and that no factor's values are statistically significantly different at an alpha level of 0.05 when `alcohol_freq` is missing. In other words, the values of `alcohol_freq` are missing likely due to some of the

original data sources not containing this variable or completely at random. It thus seemed appropriate to remove these missing values. Which left us with 69,917 rows in the dataset.

After removing the missing values, each variable was individually analyzed. Total_claims_paid is very strongly right skewed as shown in figure 2, indicating that a transformation was likely necessary, however we waited to see what model assumptions were violated before proceeding with this procedure.

The bmi has a mean of 27 and a standard deviation of 5. It is mostly normal with a slight right skew as is seen in figure 3. The smoker variable had 48,861 “never” smokers. This level was deemed as the baseline for the model and dummy variables for current (8,495 individuals) and former smokers (12,761 individuals) were created. Similarly, 45,078 individuals are “occasional” drinkers. Thus for alcohol_freq, occasional drinkers was assigned as the baseline with dummy variables encoded for weekly (19,883 individuals) and daily drinkers (5,006 individuals). Urban_rural has a mode of “urban” with 41,840 individuals. Dummy variables were thus created for suburban (17,635 individuals) and rural (10,442 individuals). For employment_status, the majority of individuals are “employed” with 47,010 occurrences. “Employed” became the baseline for the employment_status variable alongside dummy variables for unemployed (9,023 individuals) and retired (13,884 individuals). Figure 4 displays that the income variable is not normally distributed. It has a mean of \$49,907 and a standard deviation of \$46,959. Instead of performing regularization on this variable, we waited to see if the model’s assumptions were violated and remediated with transformations on Y first.

The only variables that could present a problem of multicollinearity are bmi and income. The two variables have a correlation coefficient near 0, so this is not an issue. With this assurance and the data cleaned, it seems sufficient to move forward with the next step of model selection.

Model Selection

To answer our research question, we elected to use linear regression. Linear regression allows us to predict numerical outcomes and test effects on the outcome variable. It thus is an appropriate model for determining how much individuals’ lifestyle impacts expenditures by health insurance.

The first model that was fit was a basic linear regression model with no transformations on Y. Each row in the data represents a different individual. As such, the assumption of independence is met. This model did violate the assumptions of normally distributed errors, homoscedastic errors as seen in figure 5, and a linear relationship between the variables, lack of which is displayed in figure 6.

As such, box-cox was conducted and it found that the optimal lambda for Y is 0.067. Since this is very close to zero, a log transformation was applied. A second model was fit with this log transformation on Y. This model had some very strange behavior, which was evidenced in both the QQ-plot (figure 7) and the fitted vs. residual plot (figure 8). The same three assumptions were violated, but this time two clear groups formed. Testing some data points, it became clear that the two groups were individuals who didn’t have anything spent on them and those who did.

The natural remedy to this was to try and use logistic regression instead. Logistic regression would still answer our research question, but instead of quantifying savings for the insurance company, it would give us a probability of savings. Unfortunately, the pseudo R^2 for this model is 0.00009. There also wasn't one variable that was statistically significant, so this model was useless in answering our research question as shown by figure 9.

The last model we tried was deemed "the magnitude model." This model is a linear regression, but only on the individuals who had a claim paid. This model quantifies, if a claim is paid, how does lifestyle affect the amount. This model met all of the assumptions of linear regression as seen in figures 10 and 11. As such, this was our final model.

Results

The resulting model still has fairly low predictive power. The model has an R^2 value of 0.029. This means that the model accounts for 2.9% of the variation in total_claims_paid. Figure 12 displays the fitted values vs. actual values from the model. It can be seen that there is a cloud of points with little to no pattern, which depicts the model's low predictive power. If the model had higher predictive power, we would expect to see the points coalescing around a line from the lower left hand corner to the upper right hand corner.

However, the model still has a few statistically significant coefficients. To make inference on all three simultaneously, we adjusted our confidence intervals with the Bonferroni adjustment. The three statistically significant coefficients are for smoker_current, smoker_former, and bmi.

The 95% Bonferroni adjusted family confidence intervals are: smoker_current: (\$1.52, \$1.63), smoker_former: (\$1.04, \$1.09) and bmi: (\$1.01, \$1.01). This means that we are 95% confident that the true coefficients for smoker_current, smoker_former, and bmi all lie upon their respective interval. These intervals indicate that, while BMI and smoking history do impact how much is spent by insurance companies on their subscribers, the effects are so miniscule that there *is no practical way for individuals to reduce how much their health insurance spends on them, and thereby reduce how much they spend on insurance.*

Future Research

Future research could be performed to determine how lifestyle affects total medical expenditures rather than just medical insurance expenditures. It is possible that health insurance companies are already passing down additional costs due to lifestyle onto the consumer. It would be useful to then see if lifestyle affects total medical expenditures, whether health insurance companies could adjust their plans to incentivize individuals to reduce their personal medical costs overall.

Conclusion

Mohan Thalla's, Medical Insurance Cost Prediction, dataset was cleaned and modeled to determine whether an individual's lifestyle choices affect how much health insurance spends on them and whether there is a practical way for health insurance to save and thereby pass savings down to the individual. Evidence was found that an individual's smoking history and BMI affect how much health insurance spends on them, however the amount is so miniscule that there are no practical savings that could be passed down to the health insurance subscriber.

References

Health Insurance - worldwide: Statista market forecast. (n. d.). Statista.

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<https://www.census.gov/library/publications/2024/demo/p60-284.html#:~:text=In%202023%2C%20most%20people%2C%2092.0%20percent%20or,for%20some%20or%20all%20of%20the%20year.>

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<https://www.kff.org/health-costs/2024-employer-health-benefits-survey/#e3efa8b3-48d2-458b-a2f7-c4d5add1983b--h-section-1-cost-of-health-insurance>

Thalla, M. (2025, September). *Medical Insurance Cost Prediction, Version 1*. Retrieved October 21, 2025 from

<https://www.kaggle.com/datasets/mohankrishnathalla/medical-insurance-cost-prediction>

Generative AI

ChatGPT was utilized during the coding portion of this project. It was consulted on how to efficiently perform different multilinear regression operations and for how to adjust plots.

Appendix

Figure 1

Table displaying the p-value for each test between its rows associated with the missing alcohol frequency values and the not missing values

Variable Name	P-value
Total_claims_paid (Y)	0.569 (t-test)
BMI	0.893 (t-test)
Income	0.734 (t-test)
Alcohol_freq	0.470 (Chi-sq test)
Employment_status	0.749 (Chi-sq test)
Urban_rural	0.079 (Chi-sq test)

Figure 2

Density plot of total_claims_paid

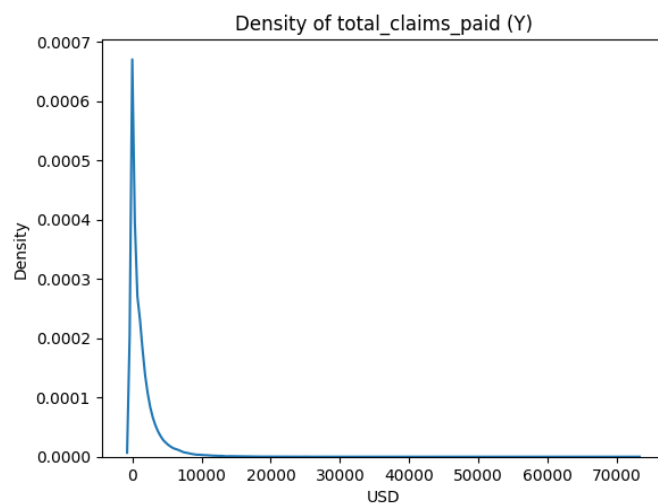


Figure 3

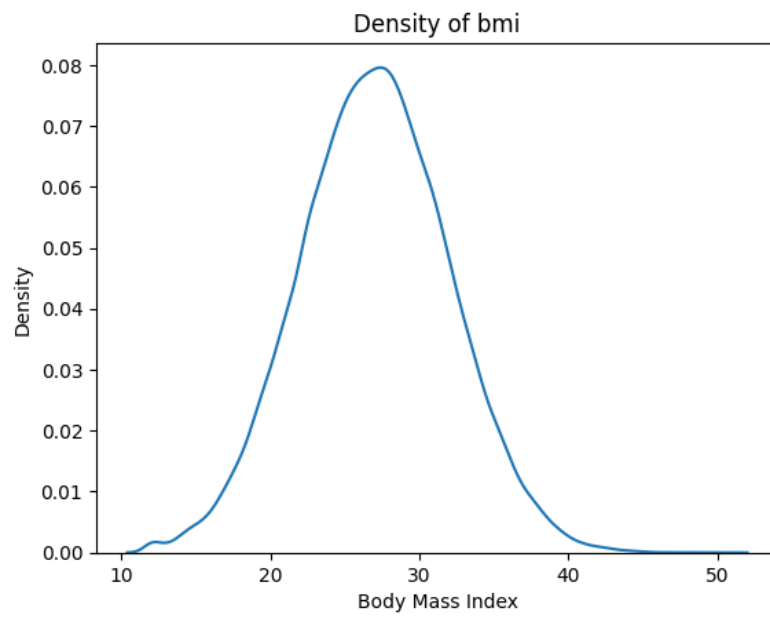
Distribution of BMI

Figure 4

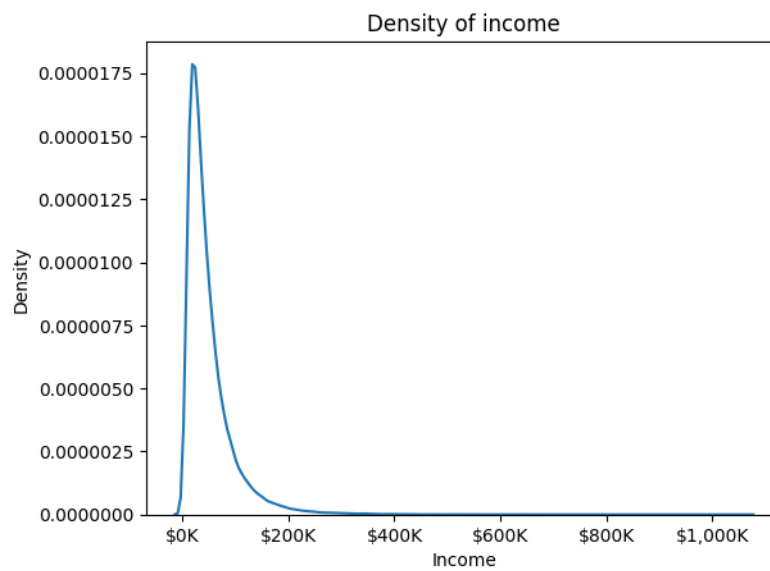
Distribution of Income

Figure 5

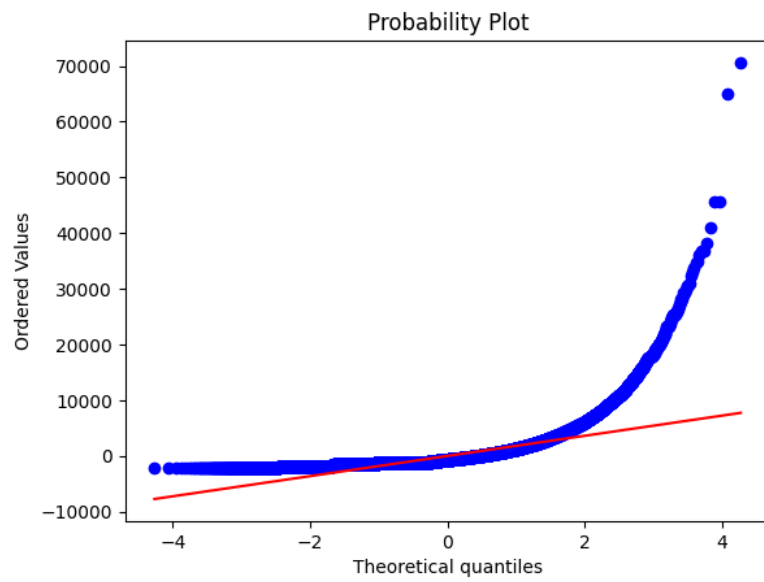
QQ-plot of the first linear regression model

Figure 6

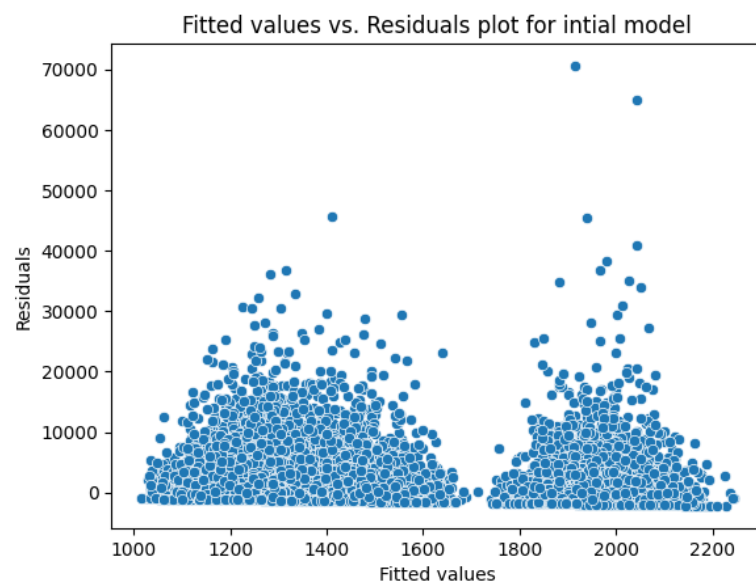
Fitted-residual plot of the first linear regression model

Figure 7

QQ-plot of the linear regression plot on $\log(y)$

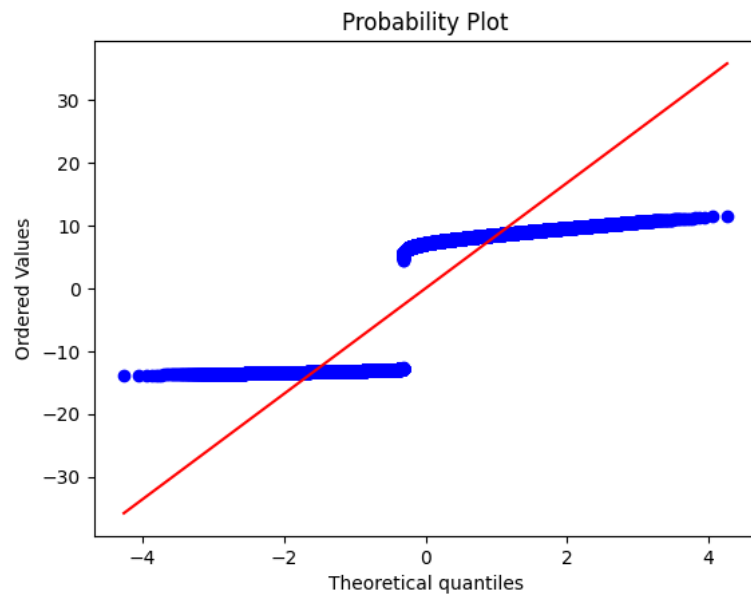


Figure 8

Fitted-residual plot of the linear regression on $\log(y)$ model

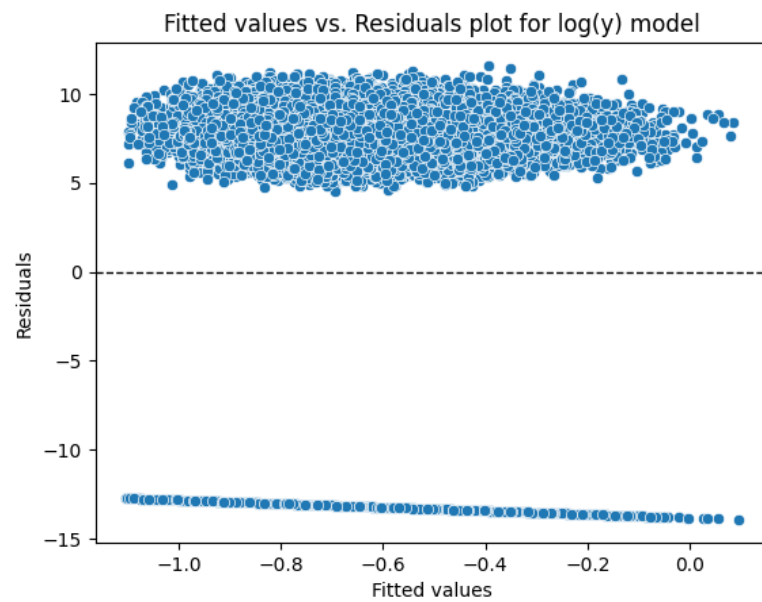


Figure 9

Logistic Regression output

Logit Regression Results						
Dep. Variable:	total_claims_paid	No. Observations:	69917			
Model:	Logit	Df Residuals:	69906			
Method:	MLE	Df Model:	10			
Date:	Sat, 08 Nov 2025	Pseudo R-squ.:	9.319e-05			
Time:	20:16:49	Log-Likelihood:	-46272.			
converged:	True	LL-Null:	-46276.			
Covariance Type:	nonrobust	LLR p-value:	0.5680			
	coef	std err	z	P> z	[0.025	0.975]
const	0.4255	0.045	9.395	0.000	0.337	0.514
income	9.757e-08	1.67e-07	0.585	0.558	-2.29e-07	4.24e-07
bmi	0.0019	0.002	1.214	0.225	-0.001	0.005
smoke_former	0.0008	0.021	0.037	0.970	-0.039	0.041
smoker_current	0.0018	0.024	0.076	0.940	-0.046	0.049
alcohol_weekly	0.0260	0.018	1.477	0.140	-0.009	0.061
alcohol_daily	0.0262	0.031	0.852	0.394	-0.034	0.087
suburban	0.0222	0.019	1.198	0.231	-0.014	0.059
rural	0.0268	0.023	1.186	0.236	-0.018	0.071
unemployed	0.0295	0.024	1.240	0.215	-0.017	0.076
retired	0.0182	0.020	0.911	0.362	-0.021	0.057

Figure 10

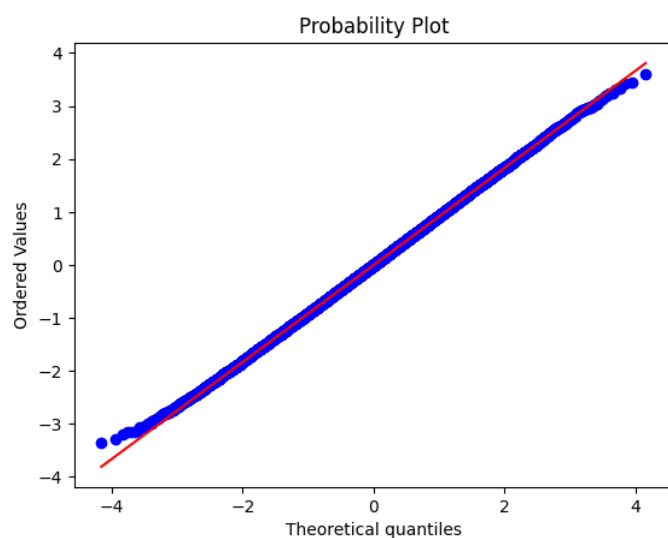
QQ-plot for final linear regression model (magnitude model)

Figure 11

Fitted-residual plot for final linear regression model (magnitude model)

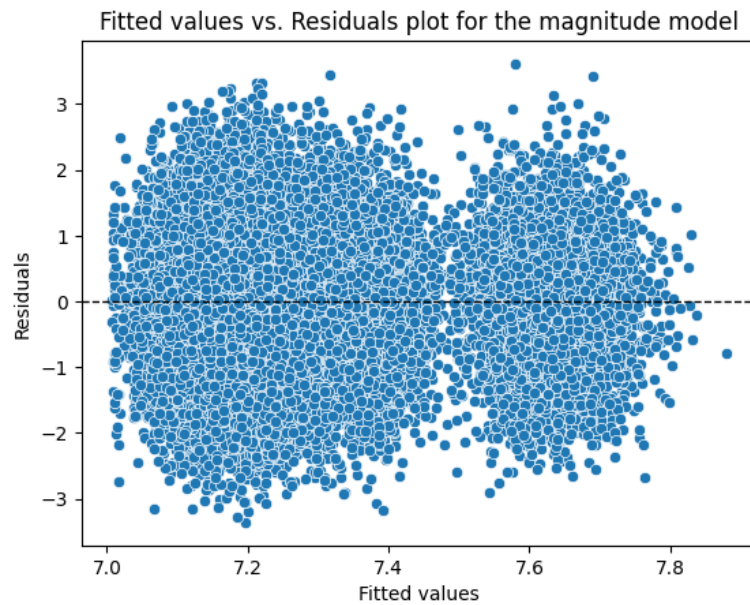
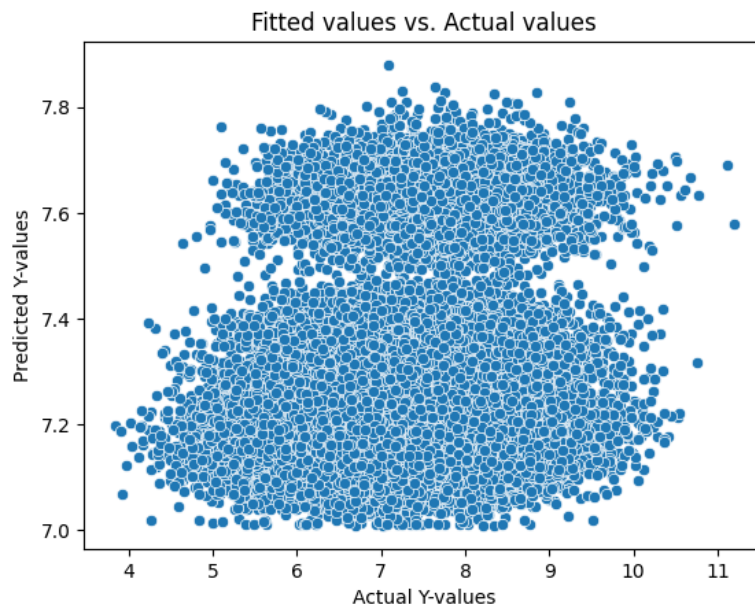


Figure 12

Fitted value vs. actual value plot of final linear regression model (magnitude model)



Bio-Sketch

James Christensen, first year M.S. Data Science student at the University of Arizona, August 2025-December 2026, contributed to the project by writing the report, identifying the data, preprocessing the data, fitting the model, and validating the model.

Peer review suggestions and actions

Nathan to James suggestions:

- numbers of slides

I took this suggestion and numbered the slides of my presentation

- use html to generate tables then copy/paste the image - sync html background color to slide color

I did not take this suggestion. I instead created my table in google slides using the functions google already has available. I then copied and pasted that table in to the appendix.

- EDA slide: state website, Data set size, IRB statement for data - single slide

I mostly took this suggestion. I had a slide with all of the information listed on a single slide. However, I did include an additional slide for EDA displaying the missing data analysis and distribution of Y

- Model slide - include equation and parameter details

I took this suggestion and created a slide that included the equation for the fitted value

- in report main body, include 1 or 2 data graphs

I didn't take this suggestion. My report was already pretty long and there was no space for any displays within it. However, the appendix does include quite a few visualizations that display the necessary aspects of the variables.